Lab01

LAB 01

THEORY

TERMINOLOGIES

Computer network
A telecommunications network where connected devices (nodes) share resources and exchange data over a data link. Connections use either cable or wireless media.

Types (by size)

- **LAN:** Local Area Network, e.g. a lab or building.
- **MAN:** Metropolitan Area Network, e.g. a city, linked through an ISP.
- **WAN:** Wide Area Network, spans countries or continents (satellite, microwave, etc.).

Host
Any device (computer or server) connected to a network. Each host has a unique identifier such as an IP address.

Topology
The arrangement of links and nodes in a network, physical or logical. The types are:

- **Bus:** all devices share one common cable.
- **Ring:** devices form a closed loop.
- **Star:** all devices connect to a central switch or hub.
- **Tree:** star networks joined in a hierarchy.
- **Mesh / fully connected:** devices link directly to many or all others.

RJ45 connector

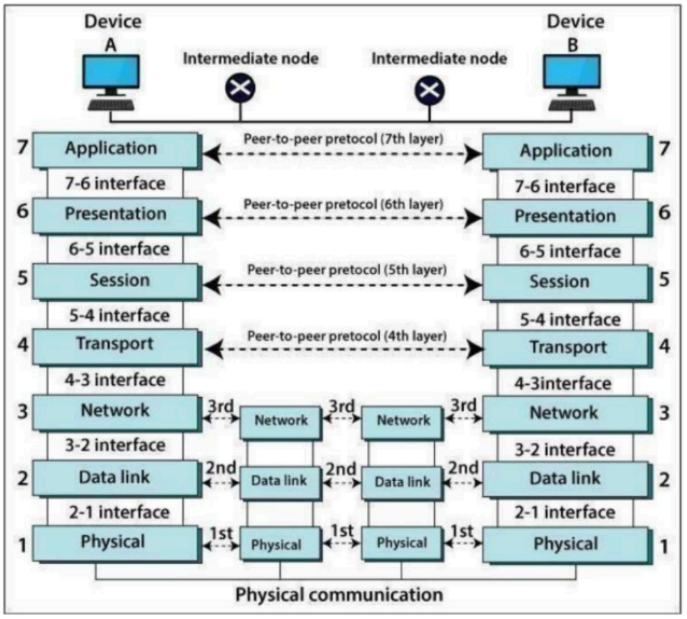
- An 8-pin, 8-position plug/jack used for Ethernet LANs.
- Two wiring standards terminate the twisted-pair cable: **T568A** and **T568B**.

TYPES OF CABLES

	Straight cable	Crossover cable
Used for	**Different** devices	**Same** type of devices
Wiring	Same standard on both ends (A to A or B to B)	T568A on one end, T568B on the other
Examples	Computer to switch, computer to modem, router WAN to modem	Computer to computer, switch to switch (normal ports), router LAN to switch normal port

OSI MODEL

A concept-based, layer-based standard for how communication systems work, aimed at interoperability. It has 7 layers, from top to bottom: Application, Presentation, Session, Transport, Network, Data Link, Physical.



COMMANDS

S.No	Linux Commands	Windows Commands	Usage/Affect
1	Ifconfig	ipconfig	To find ip address of the computer/host.
2	hostname	hostname	To display the hostname.
3	nslookup	nslookup	To list variety of info about DNS and the computers/hosts that have joined the domain.
4	Ping	ping	To check if a host can be accessed (by IP or hostname)
5	traceroute	tracert	to trace route from a host through internet router to a destination. Useful to discover why a network cannot get access to internet, and internet routing problems.
6	Netstat	Netstat	To print status of network ports, routing tables and more.

IP ADDRESS CLASSIFICATION

First octet	Class
1 to 126	A
128 to 191	B
192 to 223	C
224 to 239	D (multicast)
240 to 255	E (reserved)

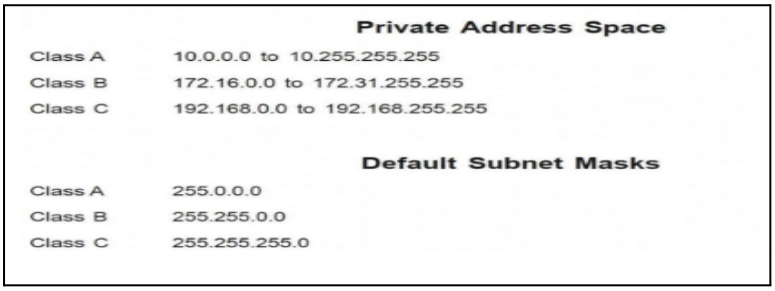
NETWORK AND HOST CLASSIFICATIONS

Class	Network portion	Host portion
A (1 to 126)	first 1 octet	last 3 octets
B (128 to 191)	first 2 octets	last 2 octets
C (192 to 223)	first 3 octets	last 1 octet

For Example

Address	Class	Network	Host
177.100.18.4	B	177.100	18.4
119.18.45.0	A	119	18.45.0
10.15.123.50	A	10	15.123.50
171.2.199.31	B	171.2	199.31

PRIVATE ADDRESS SPACE AND SUBNET MASKS



QUESTIONS

Find IP address of the computer and the IP address version also

- Command: ipconfig
- 3rd last line in output is the version and address

Find the subnet mask of the computer you currently use, MAC address, the gateway, and whether DHCP is turned on.

- Command: ipconfig /all
- Multiple networks are outputted, focus on the one you are using (eg: Wireless LAN adapter Wi-Fi)
- Physical Address is MAC address, other details are mentioned also.

Display the hostname of the computer and which protocol needs to be installed to access this command.

- Command: hostname
- Protocol: DNS or TCP/IP

By which command do we check basic IP connectivity between two computers

- Command: ping

Find out which ports on your host are connected to applications. Connect the browser to some external web page before running the appropriate command.

- Command: netstat -an or netstat -ano
- Connections showing ESTABLISHED are connected externally

Find the path of routers to a website. What is its IP address? How many hops are involved in the path?

- Command: tracert treknuces.app
- IP address and hops are also outputted

A ping to 8.8.4.4 works but a ping to the machine's name "Syrin machine" fails. What could be wrong?
DNS resolution is failing so the hostname "Syrin machine" cannot be mapped to an IP address because it is not registered in DNS or the DNS server is misconfigured.

If the order of the wires on both ends of the RJ45 connector is the same, which type of cable is used?
Straight-through cable
Can you connect a Router Ethernet port to the PC NIC using a straight-through cable? Yes
Can you connect a Switch to a PC or server using a crossover cable? No
Can you connect a Hub to a PC or server using a straight-through cable? Yes

Find all other hosts available on the network. Command: arp -a
Check any port on TCP and show the results. Command: netstat -an | findstr "TCP"
Display the number of datagrams sent and received. Command: netstat -s

Classify the following IP addresses

Address	Class
10.250.1.1	A
150.10.15.0	B
192.14.2.0	C
148.17.9.1	B
193.42.1.1	C
126.8.156.0	A
220.200.23.1	C
230.230.45.58	D
177.100.18.4	B
119.18.45.0	A
249.240.80.78	E
199.155.77.56	C
117.89.56.45	A
215.45.45.0	C
199.200.15.0	C
95.0.21.90	A
33.0.0.0	A
158.98.80.0	B
219.21.56.0	C

Classify Network and Hosts

Circle the network portion of these addresses:	Circle the host portion of these addresses:
177.100.18.4	10.15.123.50
119.18.45.0	171.2.199.31
209.240.80.78	198.125.87.177
199.155.77.56	223.250.200.222
117.89.56.45	17.45.222.45
215.45.45.0	126.201.54.231
192.200.15.0	191.41.35.112
95.0.21.90	155.25.169.227
33.0.0.0	192.15.155.2
158.98.80.0	123.102.45.254
217.21.56.0	148.17.9.155
10.250.1.1	100.25.1.1
150.10.15.0	195.0.21.98
192.14.2.0	25.250.135.46
148.17.9.1	171.102.77.77
193.42.1.1	55.250.5.5
126.8.156.0	218.155.230.14
220.200.23.1	10.250.1.1

Produce Network addresses by masking with the given IP address

Mask the corresponding 0 bits of the IP addresses

IP address	Mask	Network address
192.149.24.191	255.255.255.0	**192.149.24.0**
150.203.23.19	255.255.0.0	**150.203.0.0**
10.10.10.10	255.0.0.0	**10.0.0.0**
186.13.23.110	255.255.255.0	**186.13.23.0**
223.69.230.250	255.255.0.0	**223.69.0.0**
200.120.135.15	255.255.255.0	**200.120.135.0**

Produce Host addresses by masking with the given IP address

Mask the non corresponding 0 bits of IP addresses

IP address	Mask	Host address
222.49.49.11	255.255.255.0	**0.0.0.11**
128.23.230.19	255.255.0.0	**0.0.230.19**
10.10.10.10	255.0.0.0	**0.10.10.10**
200.113.123.11	255.255.255.0	**0.0.0.11**
223.169.23.20	255.255.0.0	**0.0.23.20**
203.20.35.215	255.255.255.0	**0.0.0.215**

Provide the Default Masks of provided IP addresses wrt their classes

IP address	Class	Default mask
103.236.162.56	A	**255.0.0.0**
162.252.172.41	B	**255.255.0.0**
180.181.68.221	B	**255.255.0.0**
203.134.40.41	C	**255.255.255.0**
211.26.68.46	C	**255.255.255.0**
17.255.253.65	A	**255.0.0.0**
1.128.96.254	A	**255.0.0.0**
108.161.213.25	A	**255.0.0.0**
12.198.43.9	A	**255.0.0.0**
211.27.133.39	C	**255.255.255.0**

Lab02

LAB 02

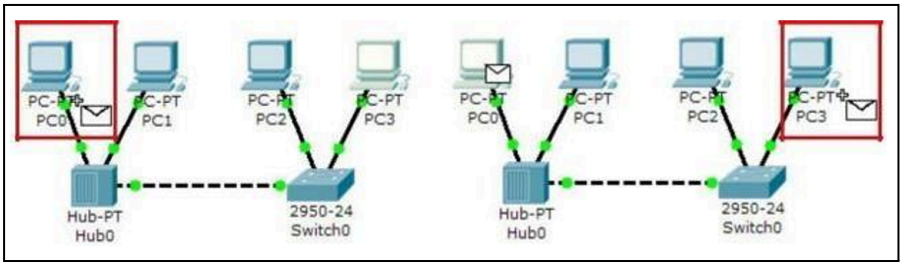
You can see how to use Cisco Packet Tracer from Lab Manual

BUILDING A CONNECTION

1. Start Packet Tracer
2. Choose Devices and Connections
   a. Select End Devices and add 3 PCs, 1 Laptop
   b. Select Connections and connect 2 PCs using copper cross-over
   c. Choose FastEthernet0 on both devices
   d. Green Arrows will be shown on the connection
3. Add a Hub (choose network devices and select it)
4. Connect Hub to End Devices using copper straight-through cable, you will see green arrows on connection
5. Add a Switch (from network devices)
6. Connect Switch like we did with hubs (select FastEthernet0/1), you will see green arrow and amber point
7. It means the switch port is temporarily not forwarding frames but goes through stages for the STP process.
8. After 30 seconds the light turns to green, indicating that the port can now forward frames.
9. Configure IP address and Subnet Masks on the hosts.
   a. Click on PC and goto Config Tab, rename PC
   b. Select FastEthernet0 below interface and enter IP address
   c. IPv4 address = 172.1.6.0 and Subnet Mask = 255.255.0.0
   d. Bandwidth and Duplex can also be changed here
   e. Repeat these steps for other PCs, using info in photo
   f. Hover over the devices after to verify the info.

Host	IP Address	Subnet Mask
PC0	172.16.1.10	255.255.0.0
PC1	172.16.1.11	255.255.0.0
PC2	172.16.1.12	255.255.0.0
PC3	172.16.1.13	255.255.0.0

10. Connect Hub to Switch using cross-over cable. You will see 1 green and 1 amber which eventually changes to green
11. Verify connectivity in real time. Also select add simple PDU tool to ping devices, also see status



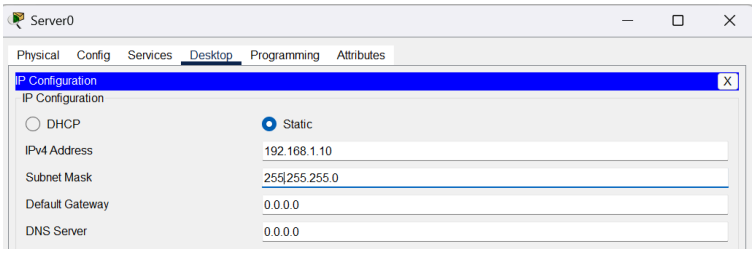
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	MyPC1	MyPC2	ICMP		0.000	N	0	(edit)	(delete)
	In Progress	MyPC4	MyPC3	ICMP		0.000	N	1	(edit)	(delete)

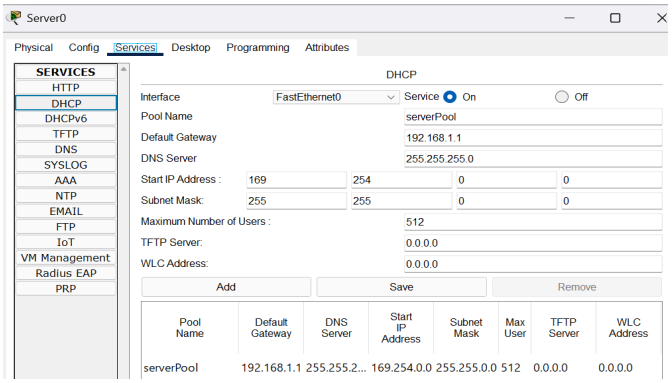
12. To Reset the network or start simulation again, click delete in PDU area, power cycle devices confirm the action and we wait for STP
13. You can save the topology in .pkt file
14. You can also simulate multiple topologies like Bus, Star, Mesh like this.

DHCP

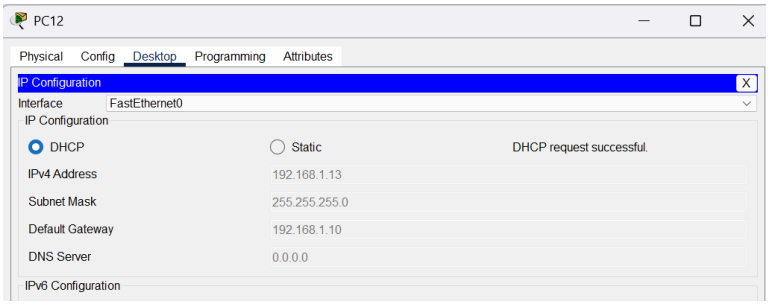
- Dynamic Host Configuration Protocol
- Network Management Protocol used for distributing IP addresses across all devices in the network.
- **DHCP Discover:** When a system is turned on in the network, it sends a request to the server. It is this.
- **DHCP Offer:** Upon receiving a request, the server sends an IP Address. It is this
- **DHCP Request:** The client responds with a packet. It is this, to eventually get the IP

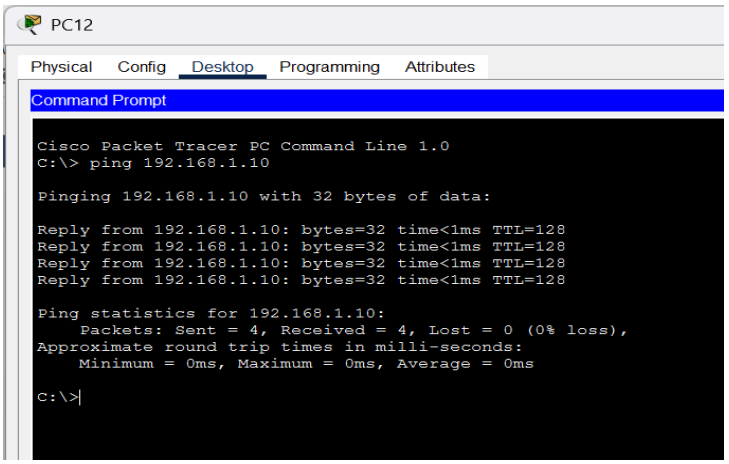
MAKING A SERVER DHCP

1. Connect multiple PCs with a switch and connect it with a server
2. Select a server, go to desktop -> IP configuration
3. 

4. Go to services tab and select DHCP
5. 

6. Save these Details
7. Select a PC and select desktop -> IP configuration, select DHCP instead of Static.
8. Request status will be shown, it should be successful.



9. Select a PC and select desktop -> command prompt
10. Ping the IP address
11. 

Lab03

LAB 03

SOCKET PROGRAMMING

- Sockets and the socket API are used to send messages across a network.
- They provide a form of interprocess communication (IPC).
- The most common type of socket applications are client-server applications.
- Sockets are the endpoints of a bidirectional communications channel.
- Sockets may be implemented over a number of different channel types: Unix domain sockets, TCP, UDP, etc.
    - **TCP**: It stands for Transmission Control Protocol, which allows for reliable communication between two applications. TCP is typically used over the Internet Protocol, which is referred to as TCP/IP.
    - **UDP**: It stands for User Datagram Protocol, a connection-less protocol that allows for packets of data to be transmitted between applications.

PYTHON

- Python provides two levels of access to network services.
- At a low level, you can access the basic socket support in the underlying operating system.
- It also provides higher-level access to specific application-level network protocols like FTP, HTTP.

ESTABLISHING A TCP CONNECTION

1. The server instantiates a ServerSocket object, denoting which port number communication is to occur on.
2. The server invokes the accept() method of the ServerSocket class. This method waits until a client connects to the server on the given port.
3. After the server is waiting, a client instantiates a Socket object, specifying the server name and port number to connect to.
4. The constructor of the Socket class attempts to connect the client to the specified server and port number. If communication is established, the client now has a Socket object capable of communicating with the server.
5. On the server side, the accept() method returns a reference to a new socket on the server that is connected to the client's socket.
6. After the connections are established, communication can occur using I/O streams.
7. The client's Output Stream is connected to the server's Input Stream, and the client's Input Stream is connected to the server's Output Stream.
8. TCP is a two-way communication protocol, so data can be sent across both streams at the same time.

SOCKET VOCABULARY

- **Domain**: The family of protocols that is used as the transport mechanism. These values are constants such asAF_INET, PF_INET, PF_UNIX, PF_X25, and so on.
- **Type**: The type of communications between the two endpoints, typically SOCK_STREAM for connection-oriented protocols and SOCK_DGRAM for connectionless protocols.
- **Protocol**: Typically, zero, this may be used to identify a variant of a protocol within a domain and type.
- **Hostname**: The identifier of a network interface.
    - A string, which can be a host name, a dotted-quad address, or an IPV6 address in colon (and possibly dot) notation.
    - A string "", which specifies an INADDR_BROADCAST address. A zero-length string, which specifies INADDR_ANY, or An Integer, interpreted as a binary address in host byte order.
- To create a socket: s = socket.socket (socket_family, socket_type, protocol=0)
    - socket_family: This is either AF_UNIX or AF_INET.
    - socket_type: This is either SOCK_STREAM or SOCK_DGRAM.
    - protocol: This is usually left out, defaulting to 0.

Once you have a socket object, then you can use required functions to create your client or server program.

SOCKET METHODS

Server Socket Methods

Method	Description
s.bind()	This method binds address (hostname, port number pair) to Socket.
s.listen()	This method sets up and start TCP listener.
s.accept()	This passively accept TCP client connection, waiting until connection arrives (blocking).

Client Socket Methods

Method	Description
s.connect()	This method actively initiates TCP server connection.

General Socket Methods

Method	Description
s.recv()	This method receives TCP message.
s.send()	This method transmits TCP message.
s.recvfrom()	This method receives UDP message.
s.sendto()	This method transmits UDP message.
s.close()	This method closes the socket.
socket.gethostname()	Returns the hostname.

EXAMPLES

```
import socket
```
[2] ✓ 0.0s

1. Getting IP Address of host

```
hostname = socket.gethostname()
ip_address = socket.gethostbyname(hostname)
print("Hostname = ", hostname)
print("IP Address = ", ip_address)
```
[4] ✓ 0.0s

```
Hostname =  Sultan-Machine
IP Address =  192.168.1.23
```

2. Getting IP Addresses of any host

```
hostname = ['google.com','treknuces.app']
for i in hostname:
    ip_address = socket.gethostbyname(i)
    print("Hostname = ", i)
    print("IP Address = ", ip_address)
```
[5] ✓ 0.1s

```
Hostname =  google.com
IP Address =  142.251.37.142
Hostname =  treknuces.app
IP Address =  216.198.79.1
```

3. Getting hostname by IP address

```
hostname1 = socket.gethostbyaddr('192.168.1.10')
hostname2 = socket.gethostbyaddr('8.8.8.8')
print("IP Address - Host 1 = ", hostname1)
print("IP Address - Host 2 = ", hostname2)
```
[8] ✓ 0.0s

```
IP Address - Host 1 =  ('192.168.1.10', [], ['192.168.1.10'])
IP Address - Host 2 =  ('dns.google', [], ['8.8.8.8'])
```

4. Getting Service name, Given port number, Protocol

```
def find_service_name():
    protocol_name = 'tcp'
    for port in [80,25]:
        servicename_tcp = socket.getservbyport(port,protocol_name)
        print(f"Port {port} -> Service Name: {servicename_tcp}")
    servicename_udp = socket.getservbyport(53,'udp')
    print(f"Port {53} -> Service Name: {servicename_udp}")

find_service_name()
```
[10] ✓ 0.0s

```
Port 80 -> Service Name: http
Port 25 -> Service Name: smtp
Port 53 -> Service Name: domain
```

5. Port Scanner

```
from socket import *
import time

startTime = time.time()

target = input("Enter Hostname to be Scanned: ")
targetIP = gethostbyname(target)
print("Scanning Host ", targetIP)

for i in range(50,500):
    s = socket(AF_INET, SOCK_STREAM)
    conn = s.connect_ex((targetIP,i))

    if (conn == 0):
        print(f"Port {i}: OPEN")
    s.close()

timeTaken = time.time()-startTime
print("Time Taken in Seconds = ", timeTaken)
print("Time Taken in Minutes = ", time.time()-startTime/60)
```
[3] ✓ 15m 14.1s

```
Scanning Host  127.0.0.1
Port 135: OPEN
Port 445: OPEN
Time Taken in Seconds =  914.0730035305023
Time Taken in Minutes =  1761082426.9179604
```

6. Client Server Connection Program

- Create 2 different files of client and server and run them to see in action

```
#Server
s = socket.socket()
print("Socket Created")
s.bind(('localhost',9999))    #binding to port
s.listen(5)     #wait for client
print("waiting for connection")

while True:
    c, addr = s.accept()    #establish connection with client
    print("Got Connection from ", addr)
    c.send(bytes("Thank You For Connecting"))
    c.close()
```

```
Socket Created
waiting for connection
```

```
#client
s = socket.socket()
print("Socket Created")
s.bind(('localhost',9999))
print(s.recv(1024).decode())
s.close()
```

Lab04

LAB 04

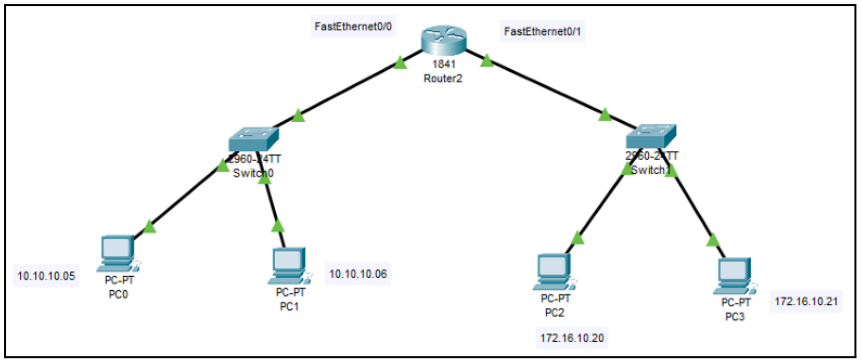
ROUTERS

A networking device that connects two or more networks together and forwards data packets between them. It directs internet traffic properly.

WORKING

- Operates at 3rd layer of OSI model
- A packet arrives and router reads its IP address
- Router checks its routing table to find best source→destination path
- It then forwards the packet to next router (hop) or destination
- This process continues until it reaches destination
- Each router has its own CPU, RAM, OS, interfaces to connect cables
- **Key Functions –** Path Determination, Packet Forwarding, Traffic Filtering, Network Segmentation

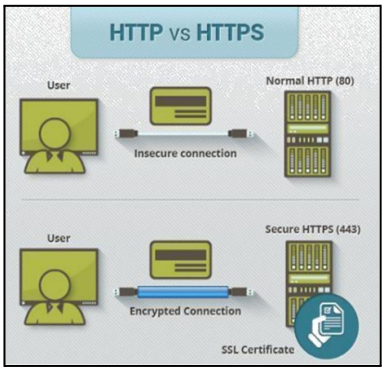
IMPLEMENTATION



First you will build the tree structure for implementing router, then configure on terminal (the router and each PC)

```
Router> enable
Router# configure terminal
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 10.10.10.1 255.0.0.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface FastEthernet0/1
Router(config-if)# ip address 172.16.10.1 255.255.0.0
Router(config-if)# no shutdown
Router(config-if)# end
```

HTTP/HTTPS



- Client requests an object from a server using HTTP protocol. Then the server responds back with the object in the form of a web page. HTTP uses port 80 for communication.
- HTTP (HTTP Secure) is a combo of 2 different protocols (HTTP and SSL/TLS). Request sending is secure and the communication is purely encrypted. HTTPS used port 443 for communication.
- A trusted public key and signed certificate is required for HTTPS.
- HTTP operates at the application layer and HTTPS at transport layer.

ERRORS

Following are client side errors:

- **400 Bad Request** – Server cannot process requests
- **401 Unauthorized** – Authentication has failed so request is forbidden
- **403 Forbidden** – Request is valid but server refuses
- **404 Not Found** – Requested resource is not available
- **408 Request Timeout** – Server timed out waiting for response

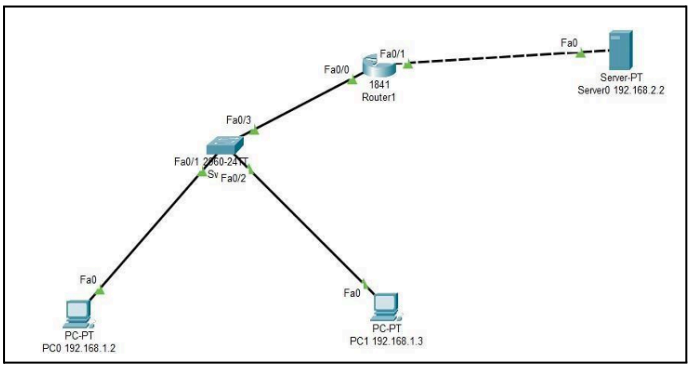
Following are server side errors:

- **500 Internal Server Error** – Generic message, when the condition is unexpected
- **501 Not Implemented** – Server does not recognize request method
- **502 Bad Gateway** – Server act as gateway or proxy and receives invalid response
- **503 Service Unavailable** – Server currently unavailable

IMPLEMENTATION

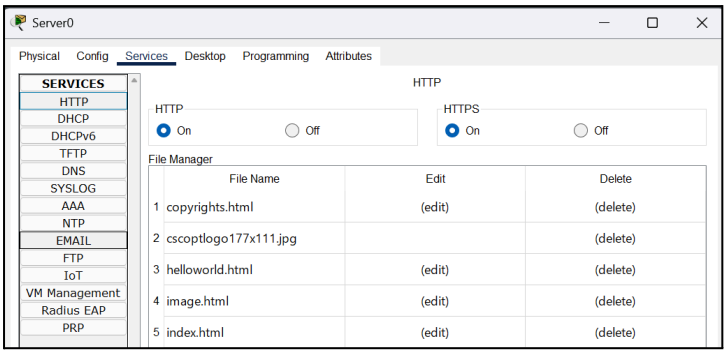
Design this topology and assign IP addresses to PCs like we did in lab2.

Then configure the router in terminal



Device	IP	Gateway
Router Fa0/0	192.168.1.1	n/a
PC0	192.168.1.10	192.168.1.1
PC1	192.168.1.11	192.168.1.1
Router Fa0/1	192.168.2.1	n/a
Server0	192.168.2.2	192.168.2.1

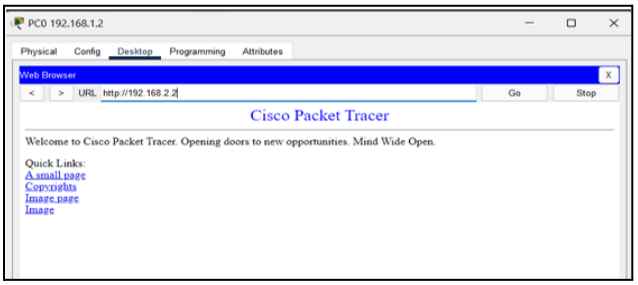
Now you will see HTTP and HTTPS services are on, by clicking on server→services→HTTP

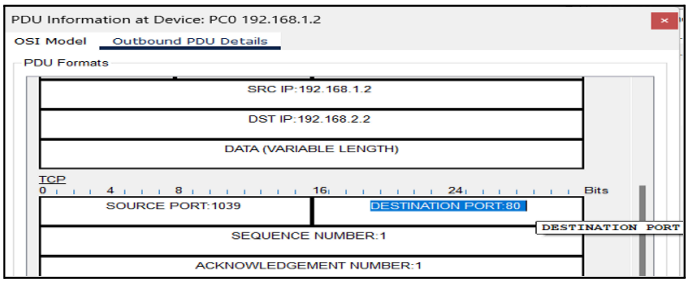


Now enter simulation mode. Edit filters and uncheck all except TCP, HTTP, ARP.

On PC0, open Desktop > Web Browser, enter http://192.168.2.2 and press Go.

Click on a Packet being transmitted to a PC and see it's destination port will be 80



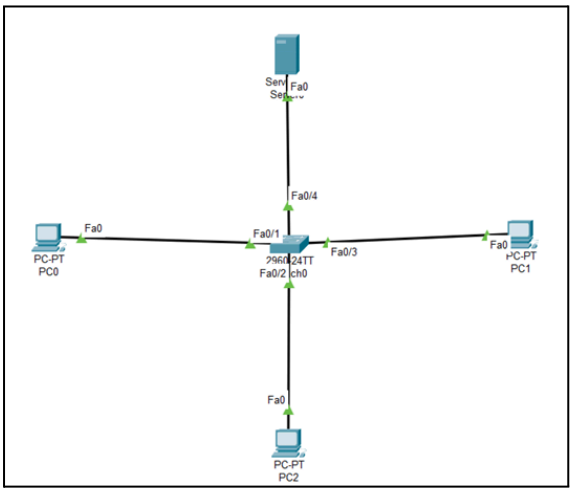


For implementing HTTPS, everything else will be the same just also check HTTPS filter and destination port will be 443.

DNS

- It is a hierarchical and distributed naming system for resources on the internet.
- It translates domain names into IP addresses for locating services.
- Common Records are : A,NS,CNAME,SOA

IMPLEMENTATION



- Implement this topology in Cisco.
- Add IP address to Server and DNS (same)
- Add IP addresses to all PCs
- The IP addresses are added in desktop -> IP config.
- Now goto server → services → DNS and turn DNS on.
- Add A type record (name fast-ai and IP address same) and CNAME record (hostname fast-ai and name fast)
- Goto web browser and type fast-ai
- Start Simulation, click on the DNS packet to see how DNS resolved the name.
- Also shows OSI layers involved.
- See Images from Manual

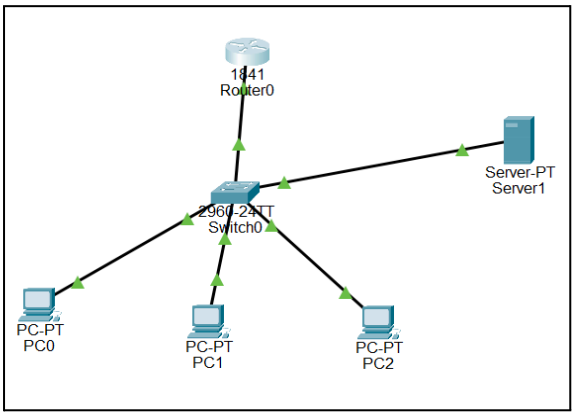
Lab05

LAB 05

SMTP

- Internet Standard for email transmission.
- User-level client mail applications typically use SMTP only for sending messages to a mail
- server for relaying.
- For retrieving messages, client applications usually use either IMAP or POP3.
- SMTP communication between mail servers uses port 25.
- Mail clients often submit outgoing emails to port 587.
- Non standard port 465 is also permitted to be used.

IMPLEMENTATION



- Make this Topology
- Configure the router (FastEthernet0 with 192.168.1.1 and FastEthernet1 with 192.168.2.1)
- Select PC→Desktop→IP configuration
- Assign the IPs along with masks and Gateways to the PCs.
- Gateway will be same on all and IP address on server will be 192.168.1.2
- IP and gateway settings for server will be server→config→FastEthernet0
- Test Connectivity: ping server IP and a 2nd PC IP

EMAIL SETUP

- Server→Services→Email
- Turn SMTP and POP3 on
- Enter domain name (fast.com) and set it
- Add users with + button by selecting id and passwords
- Configure mail client on each PC (desktop→email→configure mail)

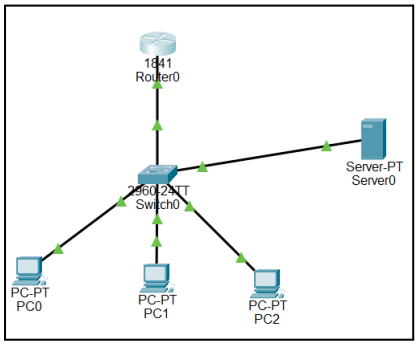
Field	PC0	Receiving PC (e.g. PC1)
Your Name	PC0	CS
Email Address	pc0@fast.gmail.com	cs@fast.gmail.com
Incoming Mail Server	192.168.1.2	192.168.1.2
Outgoing Mail Server	192.168.1.2	192.168.1.2
User Name	pc0	cs
Password	1234	1234

- Send mail from 1 PC and Receive from another email
- For Simulation, only keep TCP, POP3, SMTP filters. Click a packet and see the port address. Click Receive on the receiving PC.

FTP

- Transfers computer files between client and server on the same network.
- It is secured with SSL/TLS and uses TCP under the hood.
- It runs on 2 modes: active or passive.

IMPLEMENTATION



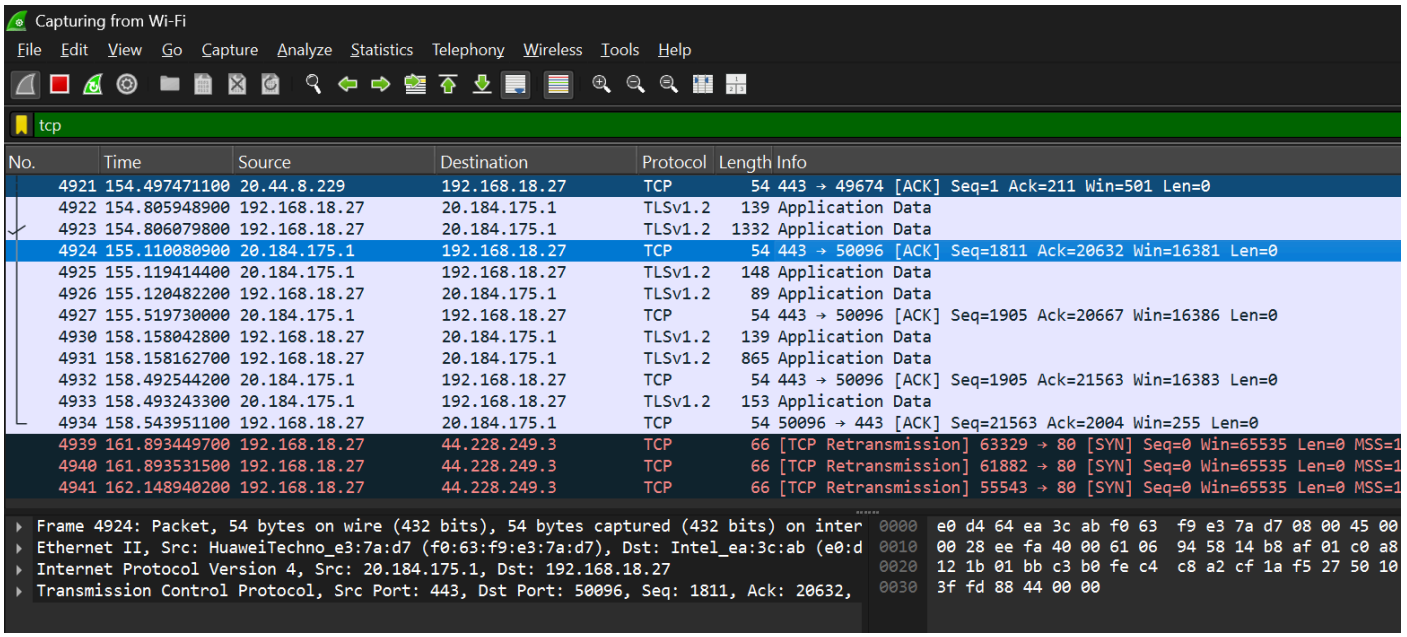
- Configure PCs, Server and Router with IPs, gateways.
- Server→Services→FTP (Turn it on)
- Create User (can use write and list permission)
- Goto PC→desktop→command prompt, connect with FTP server using username and password.

```
C:\> ftp 192.168.1.2
Trying to connect...192.168.1.2
Connected to 192.168.1.2
220- Welcome to PT Ftp server
Username:cs
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>
```

- On PC→desktop→text editor create a file
- Goto command prompt again and test it (put file.bin)
- Goto other PC's command prompt, establish connection with FTP server and write dir command to see files.
- In simulation mode, uncheck all filters except TCP, FTP. You can check packet info by clicking on it.

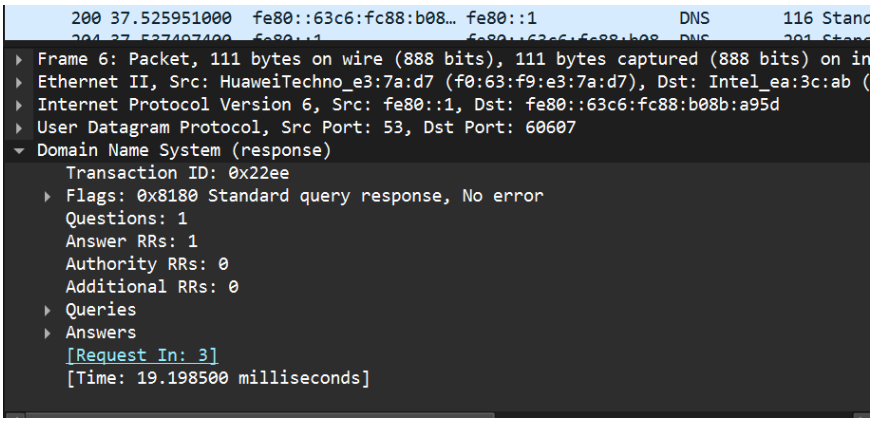
WIRESHARK

- Tool used to troubleshoot networks with performance issues.
- Install → Open → Select the network type with the moving line.
- You will see many packets. You can stop them, save them, or also filter specific packets (tcp).



QUESTIONS (WIRESHARK)

Capture DNS traffic in Wireshark and analyze the DNS request and response packets. Apply the dns Filter

- Open wireshark and select the active interface.
- In cmd, clear DNS cache (ipconfig /flushdns)
- In cmd, Generate DNS request (nslookup google.com)
- In wireshark, stop the packets and in search bar type dns
- In the info column of packets, “Standard Query” ones are requests and “Standard Query Response” are responses.
- Expand Domain Name System after selecting one of both info types
- 

Capture network traffic in Wireshark and apply an IP address filter to display packets related to a specific IP address.

- Note IPv4 address by running ipconfig in cmd.
- In the bar, write: ip.addr == 192.168.18.27
- Packets filtered.

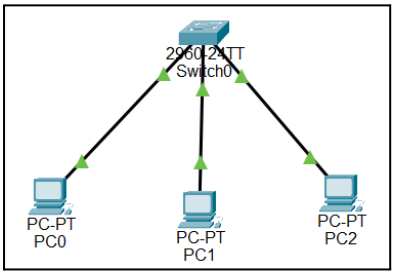
Lab06

LAB 06

TELNET

- Terminal Emulation program for TCP/IP networks.
- You enter commands on telnet and they are executed as if they were entered directly on the server console.
- This allows you to control the server and communicate with other servers on the network.

CONFIGURATION



Implement this topology and configure IPs of PCs on config→FastEthernet0.

All PCs are on vlan so we create a virtual interface on switch with vlan 1.

```
Switch>en
Switch>enable
Switch#config t
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip add
Switch(config-if)#ip address 192.168.1.2
% Incomplete command.
Switch(config-if)#ip address 192.168.1.2 255.255.255.0
Switch(config-if)#no shutdown
```

Now telnet the switch from a PC

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.1.2
Trying 192.168.1.2 ...Open

[Connection to 192.168.1.2 closed by foreign host]
C:\>
```

It refuses because we haven't applied authentication yet.

```
Switch(config-if)#exit
Switch(config)#line vty 0 15
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

—>

```
C:\>telnet 192.168.1.2
Trying 192.168.1.2 ...Open

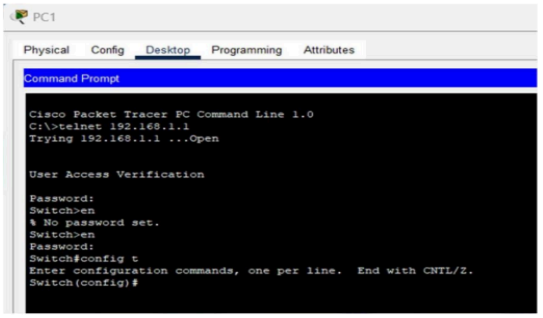

User Access Verification

Password:
Switch>
```

Now also apply password

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#enable password cs
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Now, we can go inside Switch configuration mode from our pc.



SSH

- Secure Socket Shell is a network protocol. An application layer protocol in 7th layer of OSI model.
- It supports password and key based authentication.
    - Password based: user enter username and password to authenticate to remote server.
    - Key based: authentication through key pairs (cryptographically secured keys).
- It also encrypts data communication between two devices.

CONFIGURATION

```
Switch#configure term
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Sultan-Machine
Sultan-Machine(config)#ip domain name fast-cn
Sultan-Machine(config)#crypto key generate rsa
The name for the keys will be: Sultan-Machine.fast-cn
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Sultan-Machine(config)#ip s
*Mar 1 0:15:46.715: %SSH-5-ENABLED: SSH 1.99 has been enabled
% Ambiguous command: "ip s"
Sultan-Machine(config)#ip s
Sultan-Machine(config)#ip ssh ver
Sultan-Machine(config)#ip ssh version 2
Sultan-Machine(config)#line vty 0 15
Sultan-Machine(config-line)#transport input ssh
Sultan-Machine(config-line)#
```

```
C:\>ssh -l admin 192.168.1.2

Password:

Password:



Sultan-Machine>en
Password:
Sultan-Machine#config te
Sultan-Machine#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Sultan-Machine(config)#interface FAs
Sultan-Machine(config)#interface FAstEthernet 0/3
Sultan-Machine(config-if)#no shutdown
Sultan-Machine(config-if)#exit
Sultan-Machine(config)#
```

Now, if we want to change the username from admin to something else, we will do it as follows:

```
lab01(config)#username CS-FAST secret abc
lab01(config)#line vty 0 15
lab01(config-line)#login local
lab01(config-line)#
```

Now we will do the following on our PCs:

```
C:\>ssh -l CS-FAST 192.168.1.1

Password:



lab01>
```